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Heavy Machine Days at a Chiang Rai Factory Problems, Evidence and Systematic Advice

Rev01 · 10 November 2026 · Combines: Heavy-Machine Days Report (23 Oct – 6 Nov, Rev01) · Letter on October Performance (Rev04) · VFD & Peak-Demand Advice (Rev01, 9 Nov) · ROLLER 600 machine data

The heavy machine is a **ROLLER 600 roller heat press** (380 V, 3-phase, 56 kW). About **97% of its power is heaters**; only a small motor turns the drum. The 55–65 kW peaks are therefore **heater warm-up and heater switching, not motor starts**. The fix is to control **when and how the heaters run**, not a 75 kW VFD.

Machine days (4 of 15)
70%
of all consumption and 80% of grid purchases

Peak load
58–65 kW
vs 20 kWp solar; sets the monthly demand charge

Solar share
20%
on machine days, vs 54% on normal days

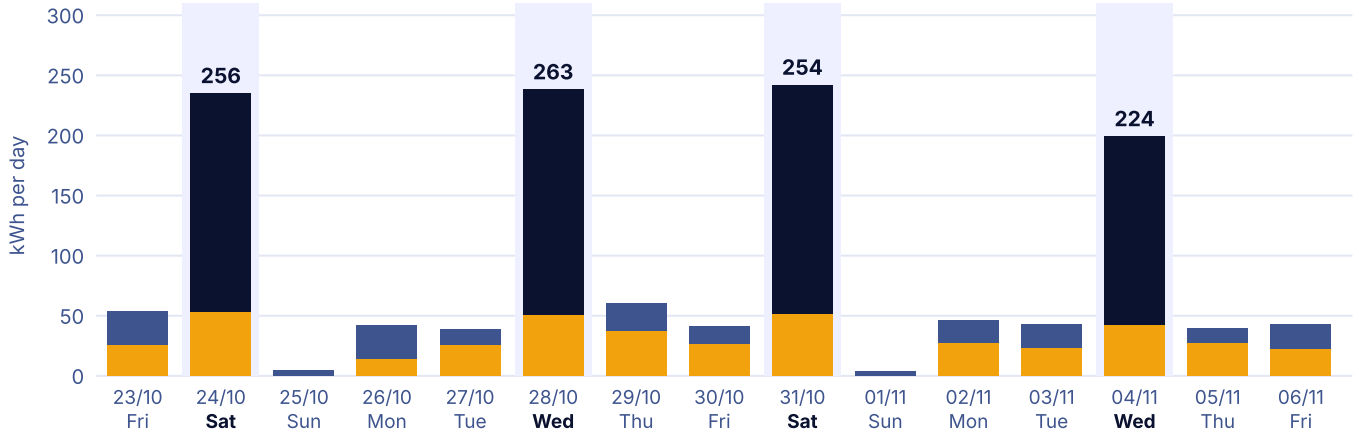
1 Problems found and the evidence

#	Problem	Evidence	Most likely cause
P1	Load jumps to 55–63 kW at 08:00–08:35	5-min data, all 4 machine days; 08:00 hour averages 38.9 kW, 34.4 kW bought; solar only 5–8 kW	Cold drum warm-up: all heaters on full power before the sun is up
P2	Repeated short peaks of 55–60 kW all day	5-min data, 24 and 28 Oct, many jumps into the evening	Heater banks switching on together (on/off temperature control), plus other loads
P3	Heavy load continues after the sun	16–23 kW through the afternoon; above 40 kW until 16:20–20:00; only 34–45% of load falls in 10:00–15:00	Drum held at temperature for long hours; work runs late
P4	High fixed part of the bill	Six PEA bills: a large fixed part plus a per-unit charge	Demand charge set by the warm-up peak (15-min average)
P5	Heat and burn damage in wiring	Burnt double connection at a main breaker (Sept); another building still double-connected	≈ 85 A continuous heater current through loose, doubled or undersized terminals
P6	Phase imbalance	Normal days: L1 carries 66% of grid purchases; usable solar ≈ 11.5 of 20 kW	Single-phase circuits mostly on L1; inverter limit ≈ 6.65 kW per phase
P7	Solar sent to the grid unpaid	554 kWh (57% of yield) in 15 days; ≈ 25 kWh even on machine days	Export allowed; machine work outside the solar window

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2 What the graphs show

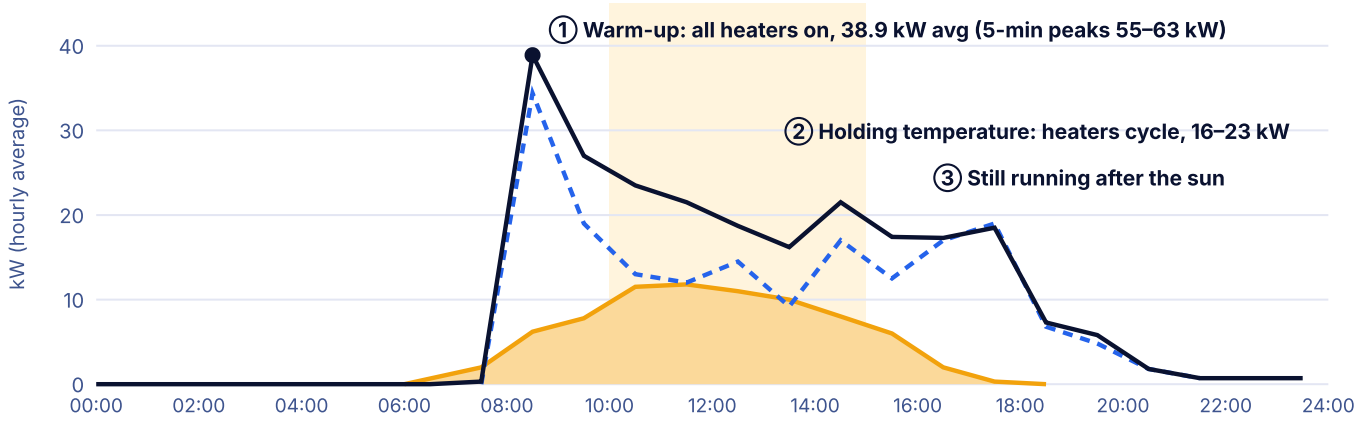
Daily: four machine days dominate the period



Solar used · Grid, machine day · Grid, other day · Machine day

Bars: solar used + bought from grid. Labels: metered consumption of each machine day. Machine days use 224–263 kWh; normal days 41–64 kWh; Sundays almost nothing. Source: daily plant data, 23 Oct – 6 Nov 2026.

Hour by hour: an average machine day



Site load · Solar output · Bought from grid (dashed)

Hourly averages of 5-minute data, machine days 23 Oct – 6 Nov. The shape fits a heat press: a warm-up spike, then a lower holding load that runs past sunset.

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3 Correction to the earlier VFD advice

Rev01 of 9 Nov assumed a 57 kW motor. With the machine identified as a 56 kW heat press:

- **No 75 kW VFD and no 55–75 kW soft starter.** Heaters have no starting surge, and a soft starter cannot set speed.
- The 610–770 A starting-current figures do not apply. Heater current is steady at ≈ 85 A when all heaters are on.
- The drum motor is small (est. 0.75–3 kW). It needs a small VFD only if the machine has no speed control already. Most do.
- Still valid from Rev01: run heavy work in the solar window, avoid overlapping loads, balance the phases, consider a battery last.

Supply
3P 380 V + N + PE
CV 4×35 mm² + G 16 mm²

→

Main MCCB
3P 100 A
Icu ≥ 25 kA

→

Heaters ≈ 53–55 kW
Temperature controller + SSR / contactors

Drum motor ≈ 0.75–3 kW
Small VFD for speed (if not fitted)

Machine supply sized for ≈ 85 A continuous (56 kW ÷ √3 ÷ 380 V). Confirm against the machine maker's manual.

4 Systematic advice in four stages

STAGE 1 · SAFETY · THIS WEEK

Make the supply safe for 85 A

- Thermal scan of the machine's cable, terminals and breaker under full load.
- Check the supply is at least CV 4×35 mm² on a 100 A MCCB; replace damaged parts.
- Re-torque all terminals; one cable per terminal.
- Correct any remaining double connection at a main breaker.

Removes the fire risk already seen on site.

STAGE 2 · CUT THE PEAK · NO COST

Change how the drum warms up

- Start warm-up at 09:30–10:00, not 08:00.
- Warm up in stages: part of the heater banks first, or a lower setpoint for the first 30–40 min (ask the machine maker how).
- No other large loads during warm-up.

Billed peak could fall from ≈ 60 kW to ≈ 30–35 kW; solar covers 8–12 kW of the warm-up.

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STAGE 3 · USE MORE SOLAR · LOW COST

Fit the work to the sun

- Batch pressing into 10:00–15:00 where orders allow.
- Heaters off, or at a lower standby temperature, in long breaks and once work ends.
- Move 1–1.5 kW of single-phase daytime circuits from L1 to L2/L3.
- Decide the export setting with Solar ESS.

Solar share on machine days up from 20%; usable solar on normal days from ≈ 11.5 toward 20 kW.

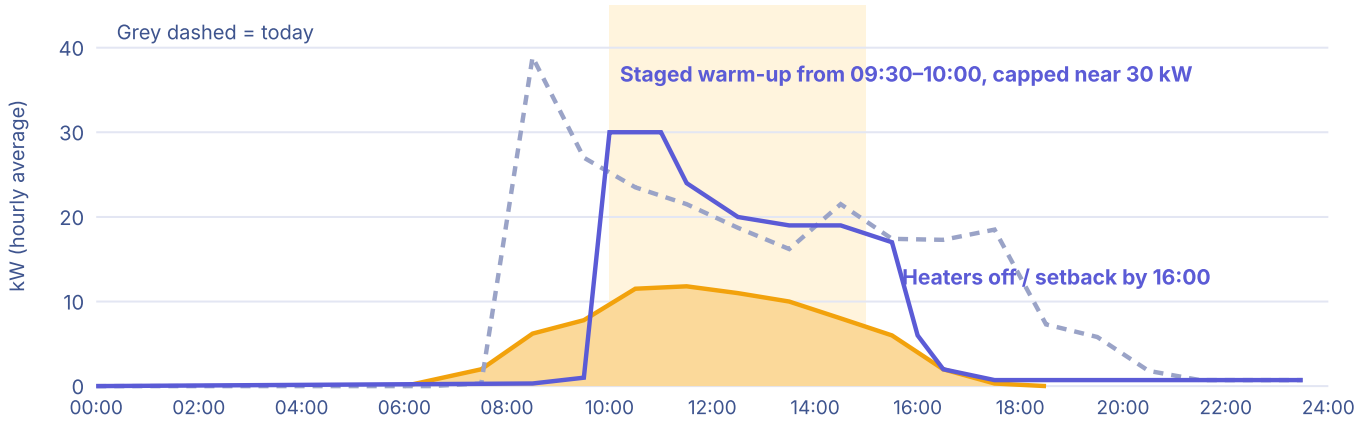
STAGE 4 · IMPROVE THE MACHINE · WITH MAKER

Smoother heater control

- Ask whether the heaters can run on PID control with SSRs, instead of on/off contactors.
- Check the drum insulation and hood; reduce standing heat loss.
- Confirm the drum motor has speed control; add a small VFD only if not.
- Consider battery peak-shaving only after Stages 1–3 are measured.

Removes the repeated 55–60 kW spikes and contactor wear.

Illustration: the same machine day after Stages 2–3



Illustrative only: same daily work, warm-up moved into the sun and capped, heaters off by 16:00. Actual results depend on the machine's heater groups and the production schedule.

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5 Expected results

Indicator	Now	After Stages 1–3	How
Monthly billed peak	≈ 58–65 kW	≈ 30–35 kW	Later, staged warm-up; no overlapping loads
Grid purchase 08:00–10:00	≈ 34 kW	≈ 0–2 kW	Machine off until 09:30–10:00
Solar share, machine day	20%	≈ 30–40%	Work moved into 10:00–15:00
Wiring temperature	Damage found	Within rating	Correct cable, breaker and terminals
Usable solar, normal day	≈ 11.5 kW	up to 20 kW	Phase balancing

Estimates from monitoring data and the published machine rating; to be confirmed by measurement after each stage.

6 Information needed to finalise

- ☐ Machine nameplate and wiring diagram
- ☐ Number and kW of heater groups; can they be switched separately?
- ☐ Temperature controller type (on/off or PID; SSR or contactor)
- ☐ Drum motor kW, and whether speed control is fitted
- ☐ Warm-up time and working setpoint
- ☐ Production schedule on Wednesdays and Saturdays
- ☐ PEA tariff type, demand rate and meter size
- ☐ Photos of the machine supply cable and breaker

7 Who does what

Party	Role
Owner	Decides the schedule, approves work on the machine and wiring, shares the information above
Owner's electrician	Stage 1 wiring work and phase balancing, to Thai EIT standards
Machine maker / service provider	Staged warm-up method, heater control, drum motor and insulation (Stage 4)
Solar ESS	Monitors results, sets the inverter and export limit, attends checks and PEA meetings, reports monthly

> Work on the machine and the building wiring is inside the owner's installation and is carried out by the owner's electrician or machine provider. Solar ESS provides technical advice and keeps the solar system tuned to the changes made.

Solar ESS Working Group